Sim-to-Real Humanoid Stand-and-Walk Control via Reinforcement Learning

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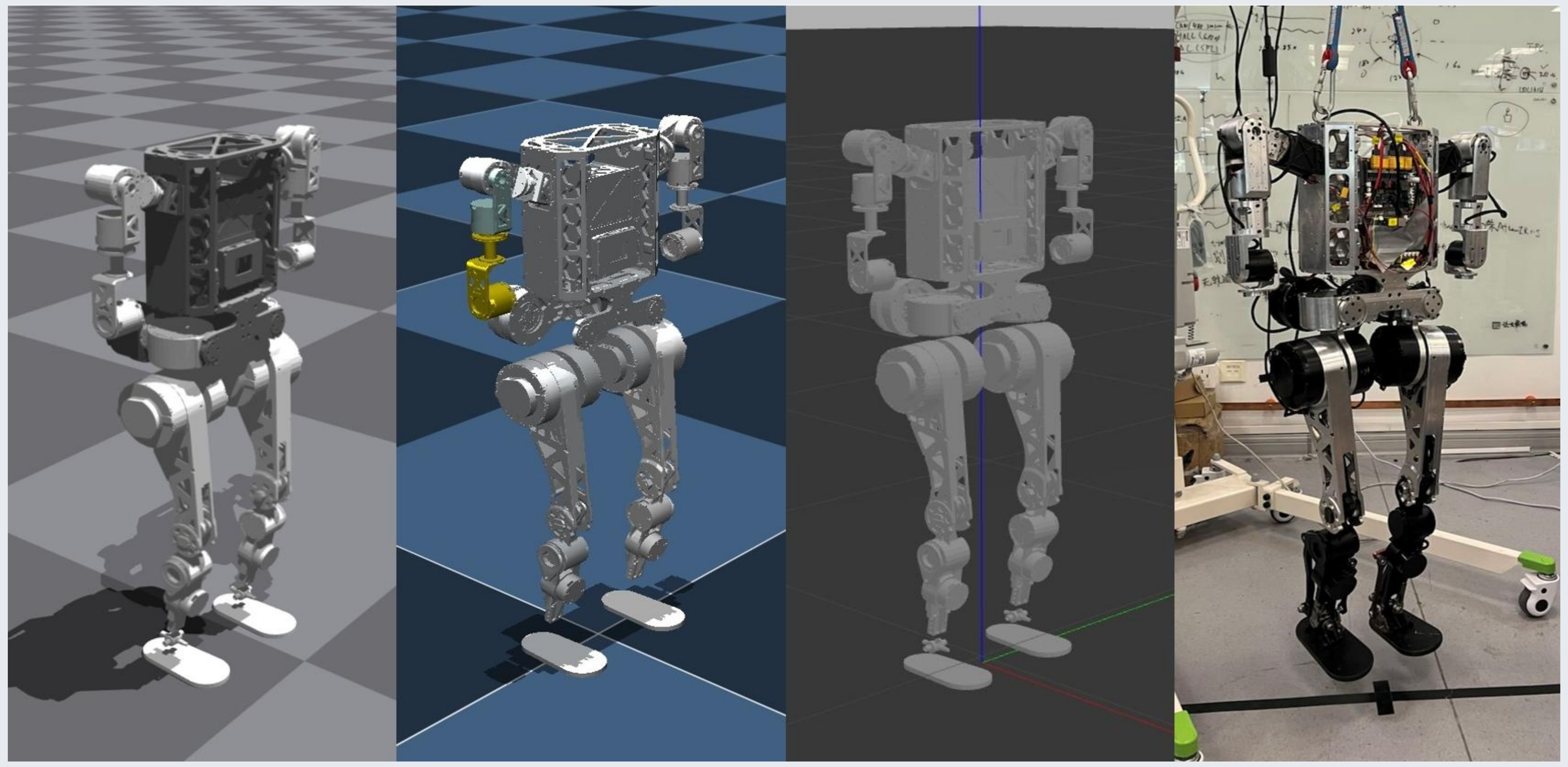
Abstract

Humanoid locomotion remains a challenge in real-world environments due to high degrees of freedom, uncertain contact dynamics, and the sim-to-real gap. This work proposes a reinforcement learning-based "Stand-and-Walk" controller for humanoid robots, supporting cross-platform training and deployment (IsaacGym, MuJoCo, Gazebo), with validation on real hardware.

Key Contributions

An internal estimator infers foot contact and base velocity, improving robustness and interpretability.

A state machine module enables natural behavior transitions, such as stopping steps when receiving a stationary command.



Problem Formulation

Problem Formulation: Asymmetric-Observation PPO for Sim-to-Real Legged Locomotion

Objective: Develop a robust walking controller for the Dora2 bipedal robot that achieves:

(1) Omnidirectional variable-speed locomotion via joystick input

(2) Physically stable stationary stance without residual oscillation

(3) An observation network capable of observing the floating base velocity and contact mask

(4) Sim-to-real transfer capability

Method

1.Partial Observability:

- Agent must infer system states (e.g., terrain interactions, floating base velocity) from limited onboard sensors
- Historical sensory integration required for state estimation

2.Sim-to-Sim and Sim-to-Real Gap:

- Dynamics discrepancies between Isaac Gym (training) and MuJoCO (validation) simulators
- Unmodeled actuator dynamics and sensor noise in physical deployment

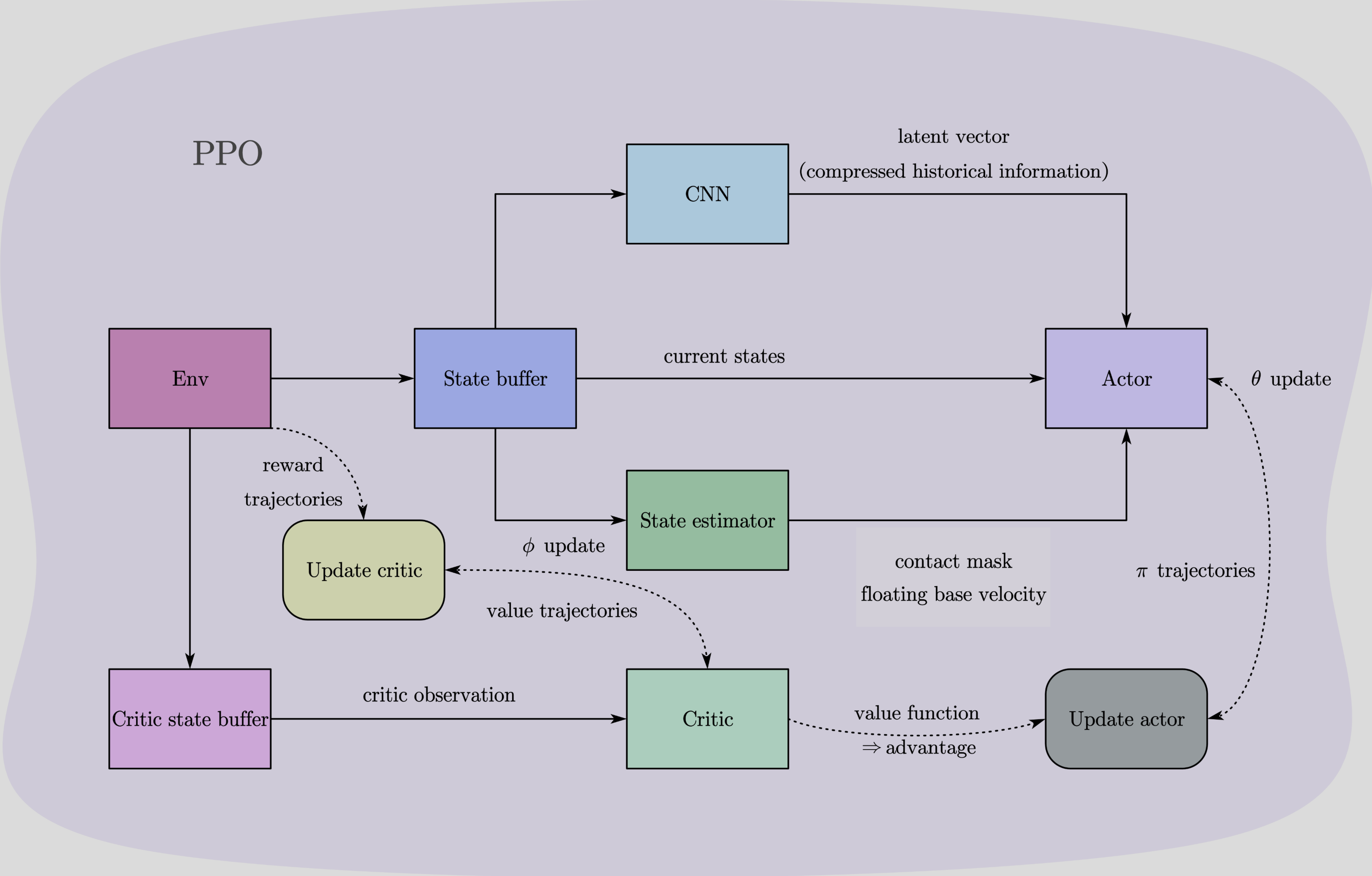
3.Observer-Augmented PPO Architecture:

- **Actor Input**:
  - Current proprioceptive state (joint positions/velocities/IMU information)
  - Observer Output: binary contact masks and floating base velocity
  - CNN-processed latent state $h_t = \mathrm{CNN}(o_{t-k:t})$ encoding historical observations
- **Critic Input**: Actor inputs + privileged information(during training)

4.Transfer Pipeline:

- Domain randomization in Isaac Gym (mass, friction, latency)
- MuJoCo and Gazebo as intermediate validation environment
- Dora2 on-board deployment

PPO-Based Humanoid Locomotion Training Framework



PPO Loss Function with Estimator Term

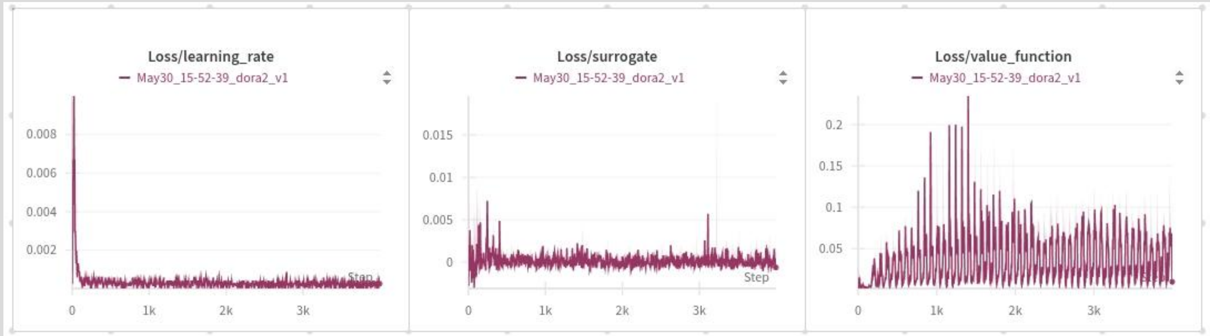
$$\mathcal{L}^{\mathrm{PPO}}(\theta) = \mathbb{E}_t\left[\mathcal{L}^{\mathrm{clip}}(\theta) - c_1 \cdot \mathcal{L}^{\mathrm{value}}(\phi) + c_2 \cdot \mathcal{L}^{\mathrm{entropy}}(\theta) + c_3 \cdot \mathcal{L}^{\mathrm{estimator}}(\theta)\right]$$

- $\mathcal{L}^{\mathrm{clip}}$ Clipped policy objective for stable learning.
- $\mathcal{L}^{\mathrm{value}}$ Value function regression loss.
- $\mathcal{L}^{\mathrm{entropy}}$ Entropy bonus to encourage exploration.
- $\mathcal{L}^{\mathrm{estimator}}$ Auxiliary supervised loss for contact and base velocity estimation:

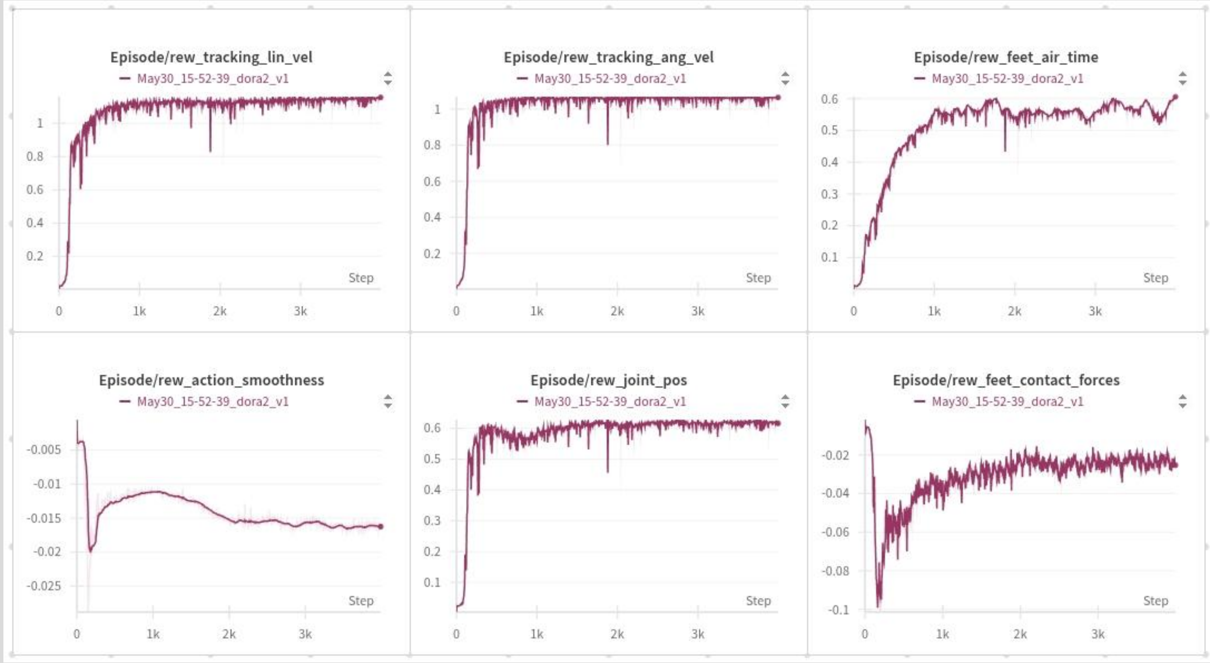
$$\left\|\hat{v}_{base} - v_{gt}\right\|^2 + \mathrm{BCE}(\hat{c}, c_{gt})$$

Experiments & Results

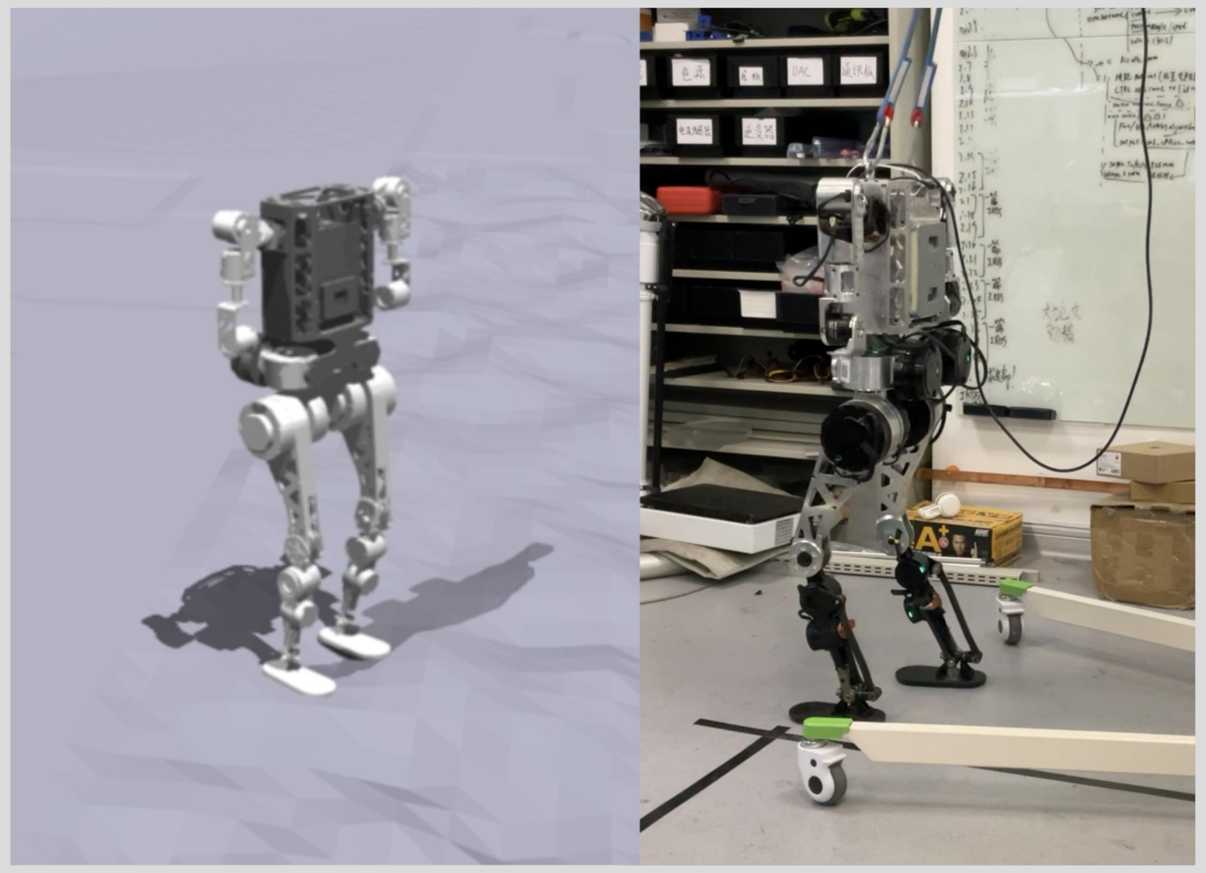
Training PPO loss



Training reward



SaW in simulation, featuring multi-terrain adaptability and real-world deployment



We propose an enhanced PPO-based reinforcement learning framework to achieve stable and autonomous locomotion for humanoid robots. By integrating a state estimator that predicts contact states and floating-base velocities, we improve the interpretability and robustness of the control policy. Our approach demonstrates strong generalization across multiple simulators, diverse terrains, and real-world deployment, achieving consistently reliable performance.